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## Unit – 1 : Session – 7 : SLO – 1 | OOP Principles

### ■ Activity (Solutions)

- 1) Encapsulation + inheritance: `class Person { private String name; public void setName(String n){ name=n; } public String getName(){ return name; } }` `class Student extends Person { int roll; }` — private field with public accessors = encapsulation; Student inherits Person's members = inheritance.
- 2) Abstraction with abstract class: `abstract class Shape { abstract double area(); }` `class Square extends Shape { double s; Square(double s){this.s=s;} double area(){ return s*s; } }` — Shape cannot be instantiated; subclasses implement the abstract method.
- 3) Polymorphism via overriding: `class Animal { void sound(){ System.out.println("some sound"); } }` `class Dog extends Animal { @Override void sound(){ System.out.println("bark"); } }` `Animal a = new Dog(); a.sound();` → prints 'bark' (run-time polymorphism).